Weather and Climate Big Data

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Hands-on Exercise 5: Extremes

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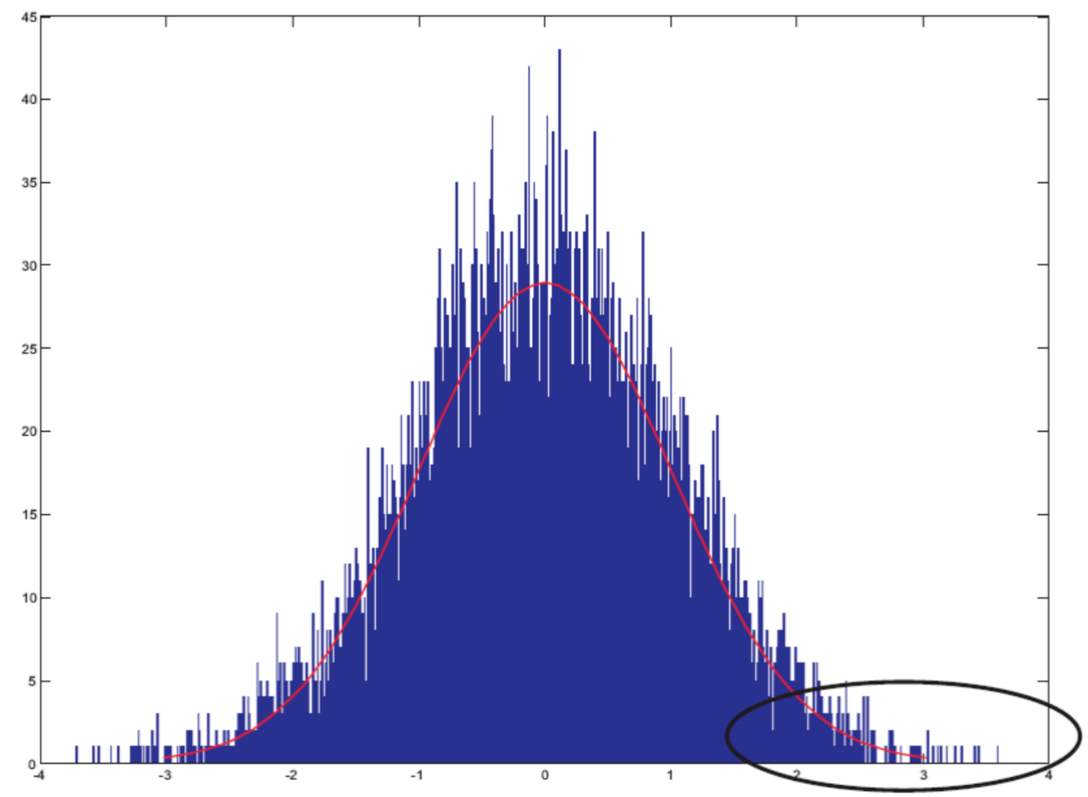
Today’s Topics

- Weather and Climate Extremes
- Compound Extremes
- Climate Attribution
- Extreme Analysis

Extreme Value Analysis

- A branch of statistics focused on rare and extreme observations
- Analyzes distribution tails, not means
- Helps estimate:
  - Return levels (e.g. 100-year flood)
  - Event probabilities (e.g. T > 40°C)



GEV distribution

- The Generalized Extreme Value (GEV) distribution is a family of continuous probability distributions developed within extreme value theory.

  - Gumbel Distribution
  - Frechet Distribution
  - Weibull Distribution

$$F(x;\mu,\sigma,\xi) = \exp\left\{-\left[1+\xi\left(\frac{x-\mu}{\sigma}\right)\right]^{-1/\xi}\right\}$$

GEV distribution

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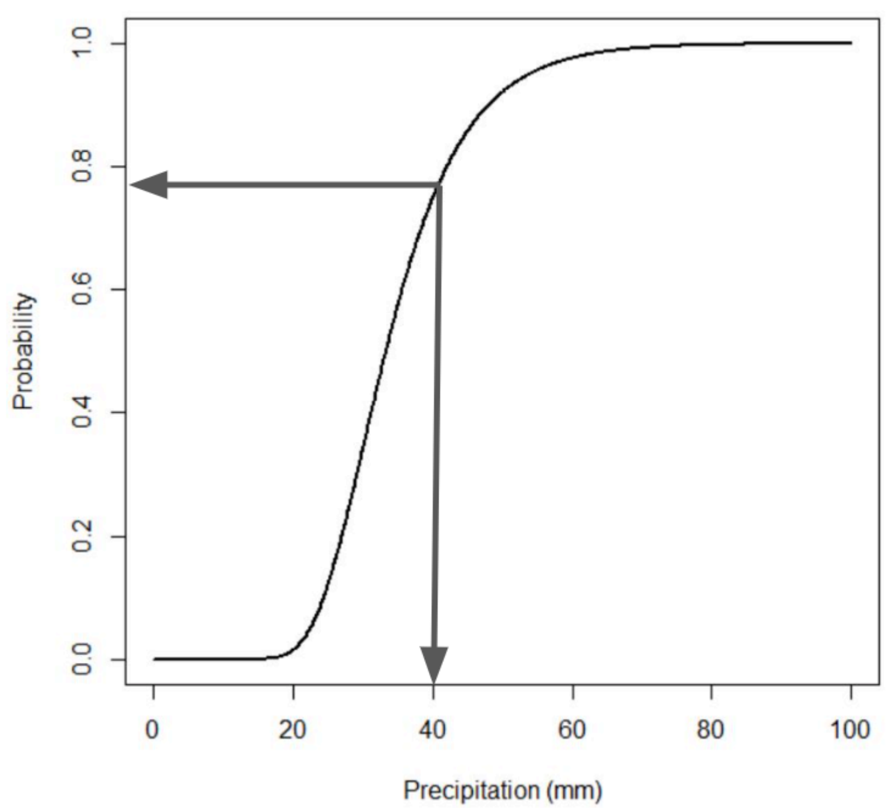
- μ: location parameter (along x-axis)
- σ: scale parameter - controls the spread
- ξ: shape parameter - affects tails

- ξ =0: Gumbel distribution
- ξ >0: Frechet distribution
- ξ <0: Weibull distribution

Return Levels

- Return level: the value x such that the probability of observing a value greater than or equal to x is 1/T (i.e. once every T years on average)
- CDF (Cumulative Distribution Function): gives the probability that a random variable is less than or equal to a given value x



For example, there is a 75% chance that an annual maximum event is less or equal to 40mm/day. Or there is a ~25% chance that an annual maximum event is +40mm/day in each year…

That means we expect to see a 40mm/day event at least once every four years... a 4-year storm event.

Return Levels

- Return level: the value x such that the probability of observing a value greater than or equal to x is 1/T (i.e. once every T years on average)
- CDF (Cumulative Distribution Function): gives the probability that a random variable is less than or equal to a given value x
- Therefore, the return level is the inverse of the CDF

$$z_T = F^{-1}(1 - \frac{1}{T})$$